

Application Deployment



Tauhid Zaman

Putting Apps Online

- We can build **full-stack apps** — a front end people click, a back end that answers
- Next step: put them **online** so other people can open a URL and use them
- You'll need a few things to do that:
 - **Git** and a **GitHub** account — so the host can pull your code
 - A **hosting platform** — a company that runs your app on the public internet (this usually costs money)



What Is Git?

- Git turns a folder into a **repository** — a project with version history
- It **tracks every change** you make to files in that folder (the green/red **diffs** on the right)
- Snapshots are called **commits** — you can go back if something breaks
- A hidden `.git` folder marks this folder as a repo and stores the **commits** that track your project's history

```

Last Turn Changes: ~ +14 -5
  ○ website/lecture/slides/lec06-csses.html +14 -5
    ~
    ▲ 177 unmodified lines
174     font-size: clamp(1.12rem, 2.4vw, 1.45rem) !important;
175     line-height: 1.35;
176   }
177
178   .skills-split-copy li > ul {
179     margin: 0.5rem 0 0;
180     padding-left: 1.15rem;
181     gap: 0.5rem;
182   }
183
184   .skills-split-copy li > ul > li {
185     font-size: clamp(1.02rem, 2.15vw, 1.28rem) !important;
186   }
187
188   .skills-split-copy a.cite-link {
189     font-size: inherit;
190     font-weight: inherit;
191   }
192
193   ~
194   ▲ 131 unmodified lines
195
196   <ul>
197     <li>We can build our class<strong>"hot phrase"</strong> full-stack app</em> – a front end people click, a back end that answers</li>
198     <li>Next step: put them <strong>online</strong> so other people can open a URL and use them</li>
199     <li>You'll need a few things to do that:</li>
200     <li><strong>Git</strong> <strong>and a</strong> <strong>GitHub</strong> account – so the host can pull your code</li>
201     <li><strong>hosting platform</strong> – a company that runs your app on the public internet (this usually costs money)</li>
202     <li>You'll need a few things to do that:
203       <ul>
204         <li><strong>Git</strong> <strong>and a</strong> <strong>GitHub</strong> account – so the host can pull your code</li>
205         <li><strong>hosting platform</strong> – a company that runs your app on the public internet (this usually costs money)</li>
206       </ul>
207     </li>
208   </ul>
209   </div>
210   <div>
211     <img

```

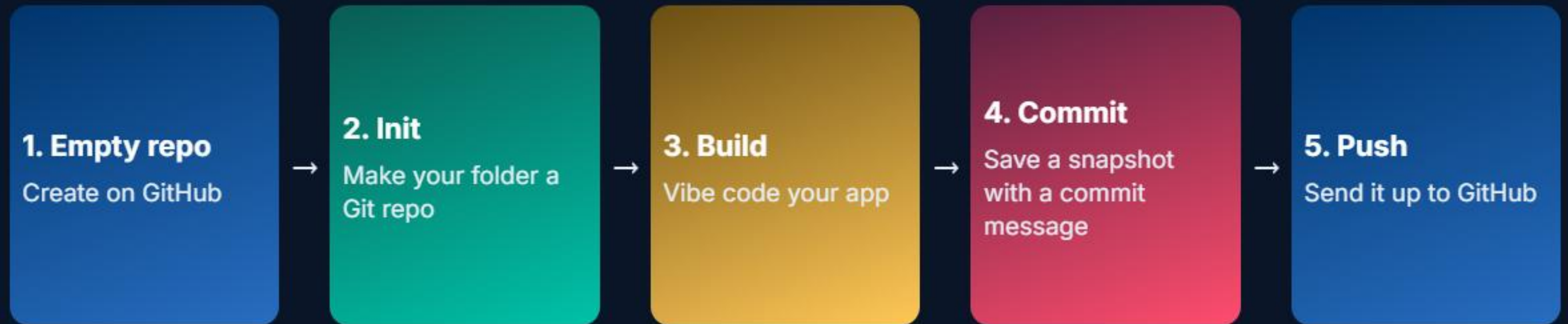
| What Is GitHub?

- **GitHub** is the website that hosts your code repositories online
- Typical flow: create an empty repo on GitHub, then **push** your local project up so it lives online
- You can also **clone** someone else's repo — download a copy to your laptop and start from their code



Pushing Code To GitHub

This is how code on your laptop gets onto GitHub



Cloning From GitHub

- Find a repo you like on the GitHub website
- **Clone** it to your machine — download a full copy you can open and run

terminal

Copy

```
git clone https://github.com/zlisto/ai_lisa_ta.git
```

AI Lisa - Teaching Assistant

A beautiful React-based chat application featuring AI Lisa, a specialized probability modeling teaching assistant. Built with OpenAI GPT-4o, MongoDB, and a modern black/pink theme.

✨ Features

- **AI-Powered Chat:** GPT-4o integration for intelligent responses
- **Image Support:** Drag & drop, paste, or upload images for analysis
- **Math Rendering:** Beautiful LaTeX math formulas with KaTeX
- **Responsive Design:** Modern UI with Tailwind CSS
- **Chat History:** Persistent storage with MongoDB
- **Vector Store Ready:** Prepared for RAG integration

🚀 Quick Start

Prerequisites

- Node.js (v16 or higher)
- MongoDB database

Vibe Coding And Git

- Once Git + GitHub are connected in your vibe coder, ask it in plain English to do the Git work
- It can:
 - Make a repo on GitHub
 - Initialize a repo in your folder
 - Commit changes
 - Push to GitHub
 - Clone a repo
- A `.gitignore` file tells Git what **not** to push to the repo

`.gitignore`

Copy

```
# secrets
.env
.env.*

# Python
.venv/
__pycache__/
*.pyc

# Node / React
node_modules/
dist/
.vite/

# local data / OS
*.db
.DS_Store
```

Databases

- A database stores many **tables**
- Tables store structured rows — users, messages, product information
- Your app **queries** the database for information
- Local database file for class today



Local Databases

- Local database files end in `.db`, e.g. `mgt409.db`
- Perfect for prototypes — the data sits in a file on your laptop, right next to your code
- `sqlite3` — Python module that lets you interact with databases



| Online Databases

- **Supabase** — an online database host for your app
- Same idea as a local `.db` — tables and queries — living on the internet so a hosted app can reach it
- Site: <https://supabase.com/>



Setting Up Supabase

1. Make a project
2. Create a database in the project — also create a database password and put it in your `.env` file
3. Get the connection string (you'll insert your password into it) and put it in your code
4. Use your vibe coder to add tables and data
5. You can also have your vibe coder connect to the Supabase MCP server (covered in a later lecture)

Connection string

Copy the connection details for your database.

 Reset database password

```
postgresql://postgres:[YOUR-PASSWORD]@db.qscblodjgsfckmcscivx.s
```

If your database password contains special characters, percent-encode them in the connection string.

Connection parameters

 Copy all

host: `db.qscblodjgsfckmcscivx.supabase.co`



port: `5432`



database: `postgres`



user: `postgres`



User Logins

- Add logins to protect your app — and to monetize it (paid accounts, subscriptions)
- User info lives in a table in your database
- Passwords must be **hashed** — turned into scrambled one-way text before you store them

users

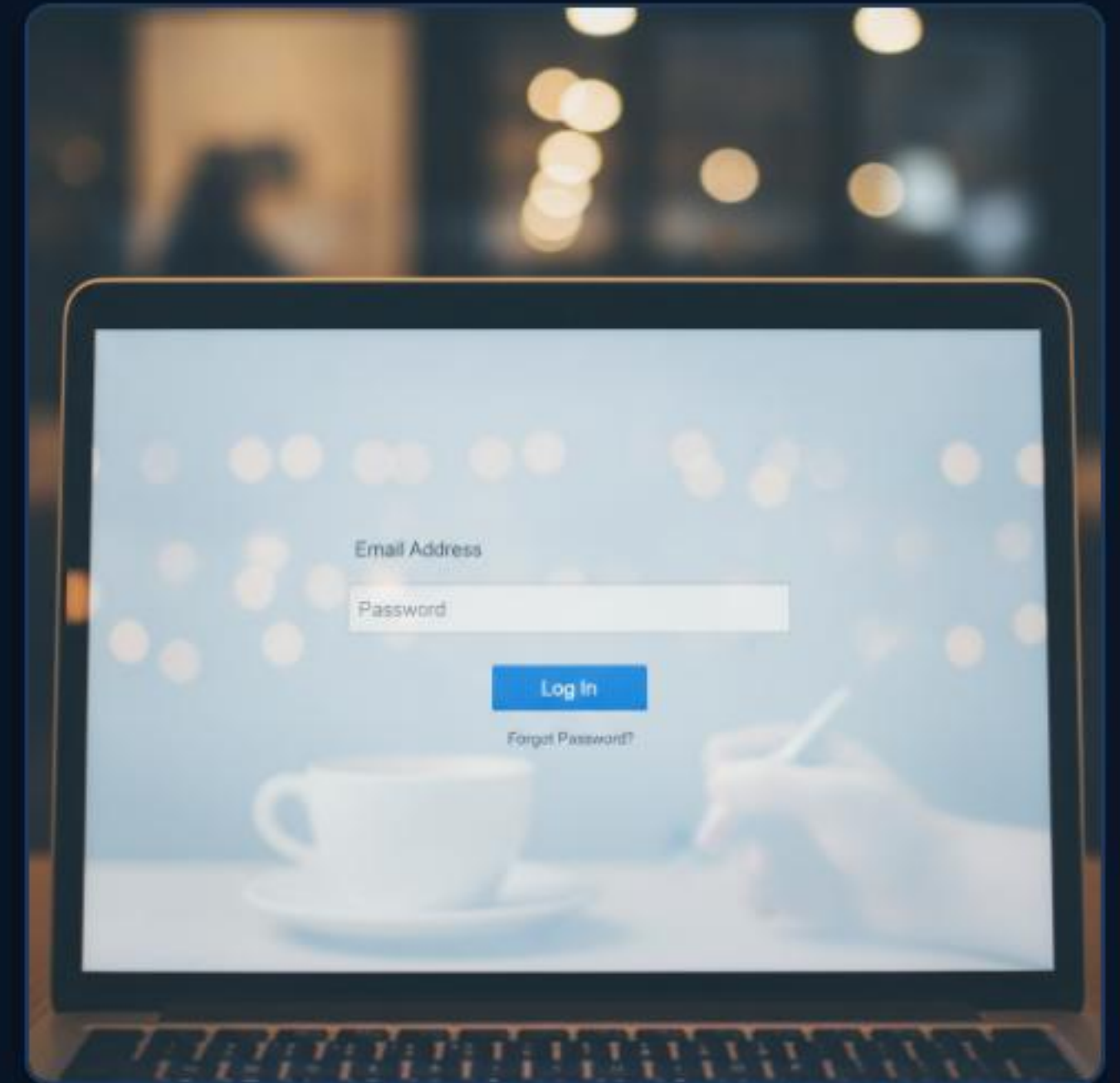
table

first_name	last_name	email	password
Maya	Chen	maya@acme.com	\$2b\$12\$eK9mQp...x7Fw
Jordan	Lee	jordan@yale.edu	\$2b\$12\$aR4nVt...9kLm
Sam	Patel	sam@startup.io	\$2b\$12\$zH2wXc...3nPq
Avery	Kim	avery@corp.com	\$2b\$12\$bN8sYd...5tRw

Password column stores a hash — scrambled one-way text

Good Password Practice

- Use a slow hash built for passwords — **bcrypt** or **PBKDF2**
 - Slow on purpose: guessing stolen hashes one-by-one takes forever
- Add a **salt** — a random string mixed into the password before hashing, so two people with the same password still get different hashes
 - In bcrypt, the salt is stored inside the password hash string — login reads it from there



Beyond A Password

- **2FA / MFA** — second factor (phone app, SMS, hardware key)
- OAuth “Sign in with Google” style logins
- Email password reset tokens, lockouts after failed logins



Putting Apps Online

- Once your app is working, the next step is to put it online
- Typical path: push code to a GitHub repo → connect a hosting service to the repo → hosting service puts the app online
- Secrets (API keys) go in the host's secure env settings



Render

- Easy to use hosting service — connect a GitHub repo and deploy
- Site: <https://render.com/>
- Can run **front end**, **back end**, and **workers**
- Typical costs
 - About **\$7/month** for a web service (backend)
 - Front end static site is free



Deploying Full Stack App On Render

- One GitHub repo → two Render services
- Deploy backend first, then front end — you need the backend URL for the front end

1. Backend — Web Service

1. Root directory: `backend/`
2. Build: `pip install -r requirements.txt`
3. Start: `uvicorn main:app --host 0.0.0.0 --port $PORT`
4. Env vars in Render (not in git): `PORTKEY_API_KEY`, `SUPABASE_DB_URL`, ...
5. You get a URL like `https://your-backend.onrender.com`

2. Frontend — Static Site

1. Root directory: `frontend/`
2. Build: `npm install && npm run build`
3. Publish directory: `dist`
4. Build env: `VITE_API_URL=https://your-backend.onrender.com`
5. You get a URL like `https://your-frontend.onrender.com`

| Other Hosting Services

- **Lovable** — strong for spinning up front ends quickly
- **AWS / Azure / Google Cloud** — full cloud platforms (more knobs)
- Pick based on what you're shipping — UI-only vs full API + workers



Lovable



Azure



Google Cloud

| Hosting Websites On GitHub

- GitHub can host a **static** website for free (HTML / CSS / JS — no backend)
- Put an `index.html` in your repo and turn on **GitHub Pages** in the repo settings
- Your site URL looks like
`https://yourusername.github.io/your-repo/`
- This class website is hosted that way:
https://zlisto.github.io/mgt_409_fa26/



Vibe Coding Session

- Update the Yale SOM Course site from Lecture 7
- Give it a database with a table of the course info
- Add a user login feature (and a `users` table in the database)
- Store user chat history in a `chats` table
- Host it online
- Vibe coding session →

